

Double Inequalities

AQA · KS4 Year 9–10

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE



$x < 5$ on a number line: open circle at 5, shade everything to the left.

COMMON MISTAKES

- Only doing the step to two of the three parts.
- Not reversing both signs when $\div$ by a negative.
- Mixing up which end is included.
- Off-by-one errors when listing integers.

METHOD

- 1 Three parts — work on all of them
- 2 Undo $\div$ across all three
- 3 Undo $\times/\div$ across all three (flip both if $\div$ negative)
- 4 Write as one statement $a < x < b$
- 5 List integers / draw the line if asked

WORKED EXAMPLE

Solve $1 < 2x + 3 \leq 9$.

Subtract 3 from all parts: $-2 < 2x \leq 6$. Divide all by 2: $-1 < x \leq 3$.

Integers: 0, 1, 2, 3.

VISUAL AID

A double inequality like $-1 < x \leq 3$ is a section of the number line: open circle at -1 (not included), filled circle at 3 (included), with the part between them shaded.

1. (2 marks)

Solve: a) $2 < x + 1 < 5$ b) $0 \leq x - 2 \leq 4$

2. (2 marks)

Solve: a) $4 < 2x < 10$ b) $6 \leq 3x \leq 18$

3. (3 marks)

Solve: a) $1 < 2x + 1 \leq 7$ b) $-3 \leq 2x - 1 < 5$ c) $5 \leq 3x + 2 \leq 14$

4. (3 marks)

List the integers: a) $-2 < x \leq 3$ b) $0 \leq x < 4$ c) $-3 \leq x \leq 1$

5. (3 marks)

Solve: a) $7 < 2x + 1 \leq 15$ b) $-4 \leq 3x - 1 < 11$ c) $2 \leq (x \div 2) + 1 < 5$

6. (3 marks)

Solve (reverse the signs): a) $1 \leq 5 - x < 4$ b) $-2 < 3 - x \leq 2$

7. (4 marks)

Solve $4 \leq 2x - 6 < 10$. a) Single inequality. b) List integers. c) Show on a number line.

8. (5 marks)

a) Length $L = 50$ cm to nearest 10 cm — write the error interval.
b) Solve $3 \leq (y+1) \div 2 < 6$ and list integer y .